GATE ME 90 Days Challenge

Day 71/90 · Practice Set with Detailed Solutions

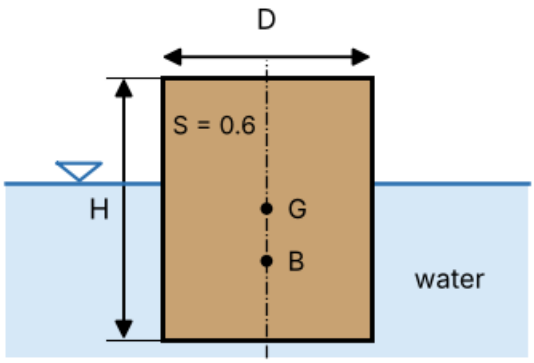
SUBJECT	TODAY'S TOPIC	COUNTDOWN
Fluid Mechanics + Heat Transfer	**FM & HT Revision**	**29 days to GATE**

How to use this set: 15 questions (8 MCQ, 7 NAT) on a full revision of Fluid Mechanics (stability of floating bodies, kinematics, radial flow and Bernoulli, angular momentum, dimensional analysis, laminar film flow, boundary-layer separation, minor losses) and Heat Transfer (conduction with heat generation, fins, Prandtl number, black-body radiation, shape factors, ε-NTU heat exchangers and the overall heat transfer coefficient). Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE. **Today's test:** open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 NAT

A solid wooden cylinder of diameter D and height H (specific gravity 0.6) floats in water with its axis vertical, as shown. The minimum ratio D/H for it to float stably in this position (round off to two decimal places) is _____



Q2 MCQ

A two-dimensional flow has the velocity field $u = x^2$, $v = -2xy$. The flow is

(A) compressible and irrotational

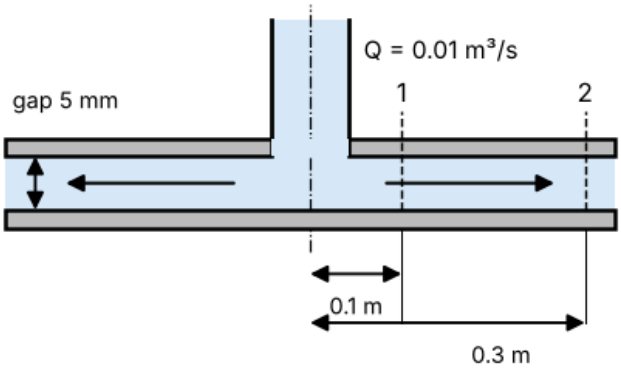
(B) compressible and rotational

(C) incompressible and rotational

(D) incompressible and irrotational

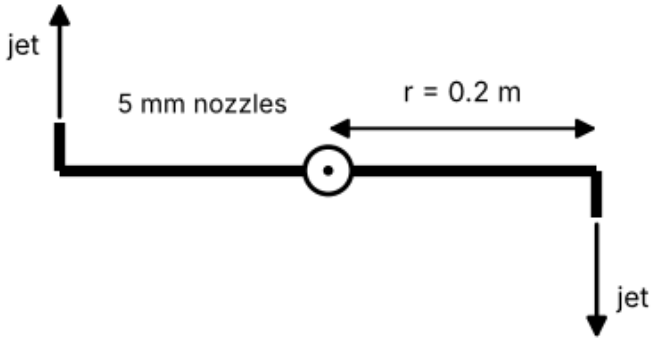
Q3 NAT

Water flows radially outward at 0.01 m³/s between two parallel horizontal discs 5 mm apart, as shown. Neglecting losses, the rise in pressure from a radius of 0.1 m to a radius of 0.3 m (in kPa, round off to two decimal places) is _____



Q4 MCQ

A lawn sprinkler, shown in plan, has two equal arms of radius 0.2 m, each ending in a 5 mm diameter nozzle pointing tangentially. The total flow is 0.4 litres per second. The torque needed to hold the sprinkler stationary is about



(A) 0.81 N·m
(B) 0.41 N·m
(C) 1.63 N·m
(D) 0.20 N·m

Q5 MCQ

The discharge Q over a triangular (V) notch depends on the head H over the notch, the acceleration due to gravity g and the notch angle θ (viscosity and surface tension neglected). Dimensional analysis gives

(A) $Q = H\sqrt{(gH)}\, f(\theta)$
(B) $Q = \sqrt{(gH)}\, f(\theta)$
(C) $Q = H^2 g\, f(\theta)$
(D) $Q = \sqrt{g}\, H^{5/2}\, f(\theta)$

Q6 NAT

Oil (density 900 kg/m³, dynamic viscosity 0.2 Pa·s) flows steadily in laminar regime as a film 3 mm thick down a wide plane inclined at 30° to the horizontal. Taking g = 9.81 m/s², the velocity at the free surface of the film (in mm/s, round off to one decimal place) is _____

Q7 MCQ

In a boundary layer on a curved surface, flow separation occurs at the point where

(A) the pressure gradient dp/dx becomes zero

(B) the velocity gradient ($\partial u/\partial y$) at the wall becomes zero in a region of adverse pressure gradient

(C) the boundary-layer thickness becomes equal to the radius of curvature

(D) the free-stream velocity reaches its maximum value

Q8 NAT

Water flows through a sudden contraction from a large pipe into a smaller pipe in which the velocity is 3 m/s. The coefficient of contraction at the vena contracta is 0.62. Taking g = 9.81 m/s², the head loss due to the contraction (in m, round off to three decimal places) is _____

Q9 MCQ

The steady temperature distribution in a plane wall 0.1 m thick (k = 40 W/m·K) is $T = 200 - 500x - 2000x^2$ (T in °C, x in m from the left face). The volumetric heat generation rate in the wall is

(A) 8.0×10^4 W/m³

(B) 1.6×10^5 W/m³

(C) 2.0×10^4 W/m³

(D) 3.6×10^5 W/m³

Q10 NAT

A long copper pin fin (k = 380 W/m·K) of 4 mm diameter has an insulated tip; h = 20 W/m²·K. The shortest fin length for which it transfers 99 % of the heat of an infinitely long fin of the same section (in mm, round off to the nearest whole number) is _____

Q11 MCQ

The Prandtl numbers of common fluids near room temperature, arranged in increasing order, are

(A) liquid mercury < air < water < engine oil

(B) air < liquid mercury < water < engine oil

(C) liquid mercury < water < air < engine oil

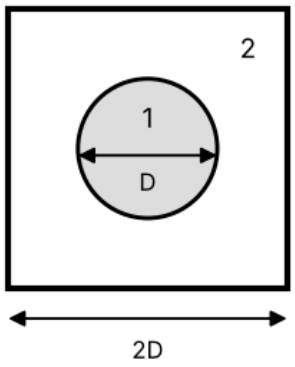
(D) engine oil < water < air < liquid mercury

Q12 NAT

A furnace whose inner walls are at 1200 K has a small circular sight-hole of 20 mm diameter in its wall. The surroundings are at 300 K. Treating the hole as a black body and taking $\sigma = 5.67 \times 10^{-8}$ W/m²·K⁴, the net rate of radiation loss through the hole (in W, round off to two decimal places) is _____

Q13 MCQ

A long cylinder of diameter D is placed centrally inside a long duct of square cross-section of side 2D, as shown. The shape factor from the inner surface of the duct to the cylinder is



(A) 0.785
(B) 0.500
(C) 0.393
(D) 0.250

Q14 NAT

In a counter-flow heat exchanger, oil (heat capacity rate 1.5 kW/K) enters at 140 °C and is cooled by water (heat capacity rate 6 kW/K) entering at 20 °C. The product UA is 2.25 kW/K. The outlet temperature of the oil (in °C, round off to one decimal place) is _____

Q15 MCQ

In a thin-walled tube heat exchanger, the inside and outside heat transfer coefficients are 1000 W/m²·K and 250 W/m²·K (wall resistance and fouling negligible). To raise the overall heat transfer coefficient, which change is more effective, and what does U become?

(A) Doubling the inside coefficient; U = 333 W/m²·K
(B) Doubling the inside coefficient; U = 222 W/m²·K
(C) Doubling the outside coefficient; U = 222 W/m²·K
(D) Doubling the outside coefficient; U = 333 W/m²·K

Answer Key

Q	Type	Answer
Q1	NAT	1.39 (accept 1.37 to 1.40)
Q2	MCQ	(C) incompressible and rotational
Q3	NAT	4.50 (accept 4.46 to 4.55)
Q4	MCQ	(A) 0.81 N·m
Q5	MCQ	(D) $Q = \sqrt{g}\, H^{5/2}\, f(\theta)$
Q6	NAT	99.3 (accept 98.5 to 100.1)
Q7	MCQ	(B) the velocity gradient ($\partial u/\partial y$) at the wall becomes zero in a region of adverse pressure gradient
Q8	NAT	0.172 (accept 0.170 to 0.174)
Q9	MCQ	(B) 1.6×10^5 W/m³
Q10	NAT	365 (accept 362 to 368)
Q11	MCQ	(A) liquid mercury < air < water < engine oil
Q12	NAT	36.79 (accept 36.50 to 37.10)
Q13	MCQ	(C) 0.393
Q14	NAT	51.8 (accept 51.3 to 52.3)
Q15	MCQ	(D) Doubling the outside coefficient; U = 333 W/m²·K

Detailed Solutions

Q1 **Answer: 1.39 (accept 1.37 to 1.40)**

Concept: A floating body is stable if its metacentric height GM = BM − BG > 0, with BM = I/V (I = second moment of the waterplane area, V = displaced volume).

Immersed depth h = 0.6H. G is at H/2 and B at h/2 = 0.3H above the base, so BG = 0.5H − 0.3H = 0.2H

BM = $(\pi D^4/64)/(\pi D^2 h/4) = D^2/(16h) = D^2/(9.6H)$

GM > 0 → $D^2/(9.6H) > 0.2H \rightarrow (D/H)^2 > 1.92 \rightarrow D/H >$ **1.39**

Exam tip: A tall thin cylinder floats on its side; it can float upright only if it is wide enough for BM to exceed BG.

Q2 **Answer: (C) incompressible and rotational**

Concept: Continuity (incompressible): $\partial u/\partial x + \partial v/\partial y = 0$. Rotation: $\omega_z = \tfrac{1}{2}(\partial v/\partial x - \partial u/\partial y)$.

$\partial u/\partial x + \partial v/\partial y = 2x - 2x = 0$: the flow is incompressible.

$\partial v/\partial x - \partial u/\partial y = -2y - 0 = -2y \neq 0$ (except on y = 0): the flow is rotational, so no velocity potential exists.

Answer: **(C) incompressible and rotational**.

Exam tip: A stream function exists for every 2-D incompressible flow; a velocity potential exists only if the flow is also irrotational.

Q3 Answer: 4.50 (accept 4.46 to 4.55)

Concept: Continuity in radial flow: $Q = 2\pi rbV$, so $V \propto 1/r$. Bernoulli (same level): $p_2 - p_1 = \tfrac{1}{2}\rho(V_1^2 - V_2^2)$.

$V_1 = 0.01/(2\pi \times 0.1 \times 0.005) = 3.183$ m/s; $V_2 = 0.01/(2\pi \times 0.3 \times 0.005) = 1.061$ m/s

$p_2 - p_1 = 500 \times (3.183^2 - 1.061^2) = 500 \times (10.132 - 1.126) = 4503$ Pa = **4.50 kPa**

Exam tip: The flow slows down as it spreads out, so the pressure rises outward (the discs are pushed apart less at the rim than near the centre).

Q4 Answer: (A) 0.81 N·m

Concept: Angular-momentum equation for a control volume: $T = \Sigma \dot{m} V r$ for the tangential jets.

Each nozzle carries 0.2 L/s: $V = 0.0002/(\pi \times 0.005^2/4) = 0.0002/1.963 \times 10^{-5} = 10.19$ m/s

$T = \rho Q_{total} V r = 1000 \times 0.0004 \times 10.19 \times 0.2 =$ **0.81 N·m (A)**

0.41 N·m counts only one nozzle.

Exam tip: If the sprinkler is free to rotate at ω, the absolute jet speed becomes $V - \omega r$, and the torque falls to zero when $\omega r = V$ (no friction).

Q5 Answer: (D) $Q = \sqrt{g}\, H^{5/2} f(\theta)$

Concept: Rayleigh's method: $Q = C\, g^a\, H^b\, f(\theta)$, θ being dimensionless.

$[L^3T^{-1}] = [LT^{-2}]^a[L]^b \rightarrow$ T: $-1 = -2a \rightarrow a = \tfrac{1}{2}$; L: $3 = a + b \rightarrow b = 5/2$

So $Q = \sqrt{g}\, H^{5/2} f(\theta)$: **(D)**. The V-notch formula $Q = (8/15)\, C_d \sqrt{(2g)} \tan(\theta/2)\, H^{5/2}$ has exactly this form.

Exam tip: Option (A) is $H^{3/2}\sqrt{g}$, the form for a rectangular notch per unit width (Q/L).

Q6 Answer: 99.3 (accept 98.5 to 100.1)

Concept: In a falling film, gravity along the plane is balanced by viscous shear; with zero shear at the free surface, $u(y) = (\rho g \sin\theta/\mu)(hy - y^2/2)$.

At $y = h$: $u_s = \rho g \sin\theta\, h^2/(2\mu)$

$u_s = 900 \times 9.81 \times 0.5 \times (0.003)^2/(2 \times 0.2) = 4414.5 \times 9 \times 10^{-6}/0.4 = 0.0993$ m/s = **99.3 mm/s**

Check: flow per metre width $q = \rho g \sin\theta\, h^3/(3\mu) = 1.99 \times 10^{-4}$ m³/s, mean velocity = 66.2 mm/s = $(2/3)u_s$; $Re = \rho Vh/\mu \approx 0.9$, laminar.

Exam tip: The film profile is half of the plane-Poiseuille parabola, so the mean is 2/3 of the surface velocity.

Q7 Answer: (B) the velocity gradient ($\partial u/\partial y$) at the wall becomes zero in a region of adverse pressure gradient

Concept: In an adverse pressure gradient ($dp/dx > 0$) the slow fluid near the wall is decelerated until it stops and reverses.

Separation point: wall shear $\tau_w = \mu(\partial u/\partial y)_{y=0} = 0$; downstream of it $(\partial u/\partial y)_{wall} < 0$ (reversed flow).

$dp/dx = 0$ marks the point of minimum pressure (maximum free-stream velocity), which lies upstream of separation: **(B)**.

Exam tip: Separation can never occur in a favourable pressure gradient ($dp/dx < 0$); a turbulent boundary layer resists separation longer than a laminar one.

Q8 **Answer: 0.172 (accept 0.170 to 0.174)**

Concept: The loss in a sudden contraction is mainly the expansion loss from the vena contracta (velocity V_2/C_c) to the full pipe: $h_L = (1/C_c - 1)^2 V_2^2/(2g)$.

$1/C_c - 1 = 1/0.62 - 1 = 0.6129$; $(0.6129)^2 = 0.3757$

$V_2^2/(2g) = 9/19.62 = 0.4587$ m

$h_L = 0.3757 \times 0.4587 =$ **0.172 m**

Exam tip: With $C_c \approx 0.62$ the loss coefficient is about 0.375; for a very large area ratio it is often rounded to $0.5V^2/2g$ (sharp entrance).

Q9 **Answer: (B) 1.6×10^5 W/m³**

Concept: Steady 1-D conduction with generation: $d^2T/dx^2 + \dot{q}/k = 0$, so $\dot{q} = -k\, d^2T/dx^2$.

$d^2T/dx^2 = -4000$ K/m² → $\dot{q} = -40 \times (-4000) =$ **1.6×10^5 W/m³ (B)**

Check: the flux leaving the right face, $-k\, dT/dx$ at $x = 0.1 = 40 \times (500 + 400) = 36\,000$ W/m², minus that entering at $x = 0$ ($40 \times 500 = 20\,000$ W/m²) $= 16\,000$ W/m² $= \dot{q}L$. ✔

Exam tip: 8.0×10^4 comes from forgetting that the second derivative of $2000x^2$ is 4000, not 2000.

Q10 **Answer: 365 (accept 362 to 368)**

Concept: Insulated-tip fin: $Q = \sqrt{(hPkA)}\, \theta_b \tanh(mL)$; infinite fin: $Q = \sqrt{(hPkA)}\, \theta_b$. Their ratio is $\tanh(mL)$.

$m = \sqrt{(hP/(kA))} = \sqrt{(4h/(kD))} = \sqrt{(4 \times 20/(380 \times 0.004))} = \sqrt{52.63} = 7.255$ m^{-1}

$\tanh(mL) = 0.99 \rightarrow mL = \tanh^{-1}(0.99) = 2.647$

$L = 2.647/7.255 = 0.3648$ m ≈ **365 mm**

Exam tip: Beyond $mL \approx 2.5$–3 extra fin length adds almost nothing, which is why copper fins are kept short.

Q11 **Answer: (A) liquid mercury < air < water < engine oil**

Concept: $Pr = \nu/\alpha$ compares momentum diffusion with heat diffusion.

Liquid metals conduct heat very well (large α): Pr ≈ 0.01–0.03 for mercury.

Gases: Pr ≈ 0.7 (air). Water: Pr ≈ 7 at 20 °C. Oils (very viscous): Pr ≈ 100–10 000.

Order: **(A) mercury < air < water < engine oil**.

Exam tip: $\delta_t/\delta \approx Pr^{-1/3}$: the thermal boundary layer is much thicker than the velocity layer for liquid metals and much thinner for oils.

Q12 **Answer: 36.79 (accept 36.50 to 37.10)**

Concept: A small opening in a large isothermal cavity absorbs almost all radiation entering it, so it behaves as a black body at the cavity temperature.

$A = \pi \times 0.02^2/4 = 3.1416 \times 10^{-4}$ m²

$Q = \sigma A(T_1^4 - T_2^4) = 5.67 \times 10^{-8} \times 3.1416 \times 10^{-4} \times (1200^4 - 300^4) = 1.7813 \times 10^{-11} \times 2.0655 \times 10^{12} =$ **36.79 W**

Exam tip: Black-body cavities are used as radiation standards; the T^4 term of the 300 K surroundings changes the answer by less than 0.4 %.

Q13 **Answer: (C) 0.393**

Concept: All radiation leaving the convex cylinder (1) strikes the duct (2): $F_{12} = 1$. Reciprocity: $A_1F_{12} = A_2F_{21}$.

Per unit length: $A_1 = \pi D$, $A_2 = 4 \times 2D = 8D$

$F_{21} = A_1/A_2 = \pi D/(8D) = \pi/8 =$ **0.393 (C)**

Exam tip: 0.785 = π/4 would result from taking the duct perimeter as 4D; always draw the geometry before using reciprocity.

Q14 **Answer: 51.8 (accept 51.3 to 52.3)**

Concept: ε-NTU method for counter flow: $\varepsilon = [1 - e^{-NTU(1-C_r)}]/[1 - C_r e^{-NTU(1-C_r)}]$.

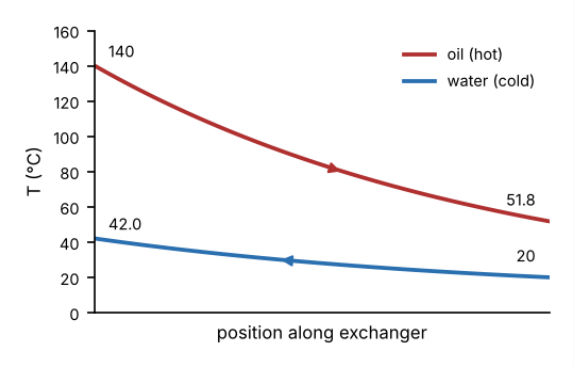
$C_{min} = 1.5$ kW/K (oil), $C_r = 1.5/6 = 0.25$, $NTU = UA/C_{min} = 2.25/1.5 = 1.5$

$e^{-1.5 \times 0.75} = e^{-1.125} = 0.3247$; $\varepsilon = (1 - 0.3247)/(1 - 0.25 \times 0.3247) = 0.6753/0.9188 = 0.7350$

$Q = \varepsilon C_{min}(T_{h,in} - T_{c,in}) = 0.7350 \times 1.5 \times 120 = 132.3$ kW

Oil outlet = $140 - 132.3/1.5 =$ **51.8 °C** (water leaves at $20 + 132.3/6 = 42.1$ °C; profiles below)

Exam tip: When outlet temperatures are unknown, use ε-NTU; LMTD would need trial and error.



Q15 **Answer: (D) Doubling the outside coefficient; U = 333 W/m²·K**

Concept: For a thin wall $1/U = 1/h_i + 1/h_o$; the larger resistance controls U.

Now: $1/U = 0.001 + 0.004 \rightarrow U = 200$ W/m²·K

Doubling h_i: $1/U = 0.0005 + 0.004 \rightarrow U = 222$ W/m²·K

Doubling h_o: $1/U = 0.001 + 0.002 \rightarrow U = 333$ W/m²·K: **(D)**

Exam tip: Always attack the side with the lower h (usually the gas side); that is also where fins are added.

What next?

Day 71 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

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Next, Day 72/90: Full-Length Mock 1. Comment **GATE90** on our reel to get all links.
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